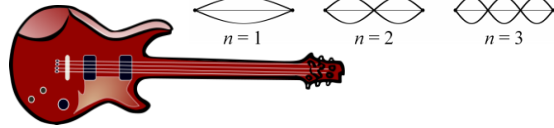


Fourier series example: guitar string



$$\psi(x, t) = A^+ e^{i(\omega t - kx)} + A^- e^{i(\omega t + kx)}$$

$x = 0$
 $x = L$

Boundary conditions:

$$0 = \psi(x = 0, t) = A^+ e^{i(\omega t)} + A^- e^{i(\omega t)} \Rightarrow A^- = -A^+$$

$$0 = \psi(x = L, t) = A^+ (e^{i(\omega t - kL)} - e^{i(\omega t + kL)}) = A^+ e^{i\omega t} (e^{-ikL} - e^{ikL}) = -A^+ e^{i\omega t} \sin(kL) = 0$$

$$\Rightarrow kL = n \cdot \pi, \quad n = 0, 1, 2, 3, \dots$$

The quantization of the k wavenumber also yields the quantization of the frequency due to the $\omega/k = v_{\text{wave}}$ condition. Accordingly, the wavefunction can include a standing wave with the base frequency, as well as higher harmonics:

$$\psi(x, t) = \sum_n A_n \sin(k_n x) \cos(\omega_n t + \alpha_n), \quad k_n = n \frac{\pi}{L}, \quad \omega_n = n \frac{\pi v_{\text{wave}}}{L}, \quad n = 1, 2, 3, \dots$$

The time-dependent part remains periodic with $T = 2L/v_{\text{wave}}$ period time even after adding an arbitrary number of harmonics.

Reminder: different instruments played at the same musical frequency (fundamental frequency) have different tone colors because the different harmonics of the fundamental frequency. The tone color is attributed to the different amplitudes and phases of the higher harmonic components.

1

Fourier series

Statement: an arbitrary periodic function (for which $F(t+T)=F(t)$ for all t) can be expressed as the sum of harmonic functions including a DC component, a sine and cosine component with the base frequency, $f_1=1/T$ ($\omega_1=2\pi f_1$) and infinite sine and cosine components with the subharmonics of the base frequency, $f_n=n \cdot f_1$, $n=2, 3, 4, \dots$:

$$F(t) = \sum_{n=0}^{\infty} A_n^{\sin} \sin(\omega_n t) + A_n^{\cos} \cos(\omega_n t), \quad \omega_n = n \frac{2\pi}{T}, \quad n = 0, 1, 2, 3, \dots$$

Alternatively, one can use a complex notation. In this case the coefficient a_n is complex, the real part corresponds to the sine term, while the complex part corresponds to the cosine term. More precisely:

$$F(t) = \text{Re} \sum_{n=0}^{\infty} a_n \exp(i\omega_n t), \quad \omega_n = n \frac{2\pi}{T} = n \cdot 2\pi \cdot f_1, \quad n = 0, 1, 2, 3, \dots \quad a_n = A_n^{\cos} - iA_n^{\sin}$$

Calculating the complex coefficients (note, that for the $\omega_0=0$ DC term the sine component is anyhow zero, $\sin(\omega_0 t)=0$, i.e. a_0 can be taken to be real without the loss of generality!):

$$F(t) = \text{Re} \sum_{n=0}^{\infty} a_n \exp(i\omega_n t) = \frac{1}{2} \sum_{n=0}^{\infty} (a_n \exp(i\omega_n t) + a_n^* \exp(-i\omega_n t))$$

Let's multiply both sides by $\exp(-i\omega_m t)$ and integrate for an entire period time, T :

$$\int_{t_0}^{t_0+T} F(t) e^{-i\omega_m t} dt = \frac{1}{2} \sum_{n=0}^{\infty} a_n \int_{t_0}^{t_0+T} e^{i(\omega_n - \omega_m)t} dt + a_n^* \int_{t_0}^{t_0+T} e^{i(-\omega_n - \omega_m)t} dt = \begin{cases} T a_m / 2, & m > 0 \\ T a_0, & m = 0 \end{cases} \Rightarrow$$

$$a_m = \frac{2}{T} \int_{t_0}^{t_0+T} F(t) e^{-i\omega_m t} dt$$

$$a_0 = \frac{1}{T} \int_{t_0}^{t_0+T} F(t) dt$$

$\omega_m = m \frac{2\pi}{T}, \quad m = 0, 1, 2, 3, \dots$

This is how the Fourier coefficients are calculated

2

Examples of Fourier decomposition

Sawtooth signal

the first period is:

$$F(t, t = 0..T) = -\frac{A}{2} + \frac{A}{T} \cdot t$$

Fourier coefficients:

$$a_0 = \frac{1}{T} \int_0^T F(t) dt = 0 \quad \int t e^{at} = \frac{e^{at}}{a^2} (at - 1)$$

$$a_m = \frac{2}{T} \int_0^T F(t) e^{-i\omega_m t} dt = \frac{2A}{T^2} \int_0^T t e^{-i\omega_m t} dt = \frac{2A}{T^2} \left[\frac{e^{-i\omega_m t}}{\omega_m^2} (i\omega_m t + 1) \right]_0^T = \frac{2A}{T^2} \left[\frac{e^{-i2\pi m}}{\left(\frac{2\pi m}{T}\right)^2} (i2\pi m + 1) - \frac{1}{\left(\frac{2\pi m}{T}\right)^2} \right] = \frac{iA}{\pi m}$$

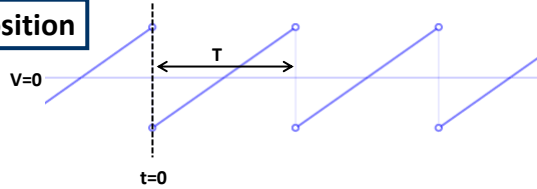
$$F(t) = -\frac{A}{\pi} \sum_{n=1}^{\infty} \frac{\sin(2\pi n t/T)}{n}$$

Symmetric triangle signal:

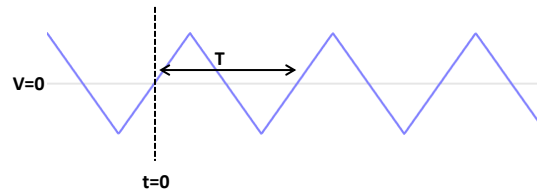
$$F(t) = \frac{8A}{\pi^2} \sum_{n=0}^{\infty} (-1)^n \cdot \frac{\sin(2\pi(2n+1)t/T)}{(2n+1)^2}$$

The even subharmonics are missing from symmetry reasons

Since there is no sudden change, it can be well approximated without high harmonics.



Experiment: sound and FFT of an asymmetric triangle signal



3

Fourier transformation

A function of period T can be described by Fourier decomposition with discrete components of circular frequencies spaced $\Delta\omega = 2\pi/T$ apart. For $T \rightarrow \infty$ $\Delta\omega \rightarrow 0$, i.e. any frequency ω can contribute to the function $F(t)$.

Definition: the Fourier transform of an $F(t)$

(not necessarily periodic) function:

(notation: upper case - time space, lower case - circular frequency space)

$$f(\omega) = \mathcal{F}(F(t)) = \int_{-\infty}^{\infty} F(t) e^{-i\omega t} dt$$

Example:

$$F(t) = e^{i\omega_0 t} \Rightarrow f(\omega) = \int_{-\infty}^{\infty} e^{i(\omega_0 - \omega)t} dt = \begin{cases} 0, & \text{if } \omega_0 \neq \omega \\ \infty, & \text{if } \omega_0 = \omega \end{cases} \stackrel{\text{def}}{=} 2\pi \delta(\omega_0 - \omega)$$

Definition: the inverse Fourier transform of the $f(\omega)$ function: $\mathcal{F}^{-1}(f(\omega)) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(\omega) e^{i\omega t} d\omega$

Fourier integral theorem (approximate): the inverse Fourier transform of the Fourier transform of a function returns the original function: $F(t) = \mathcal{F}^{-1}(\mathcal{F}(F(t)))$

$$F(t') = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega e^{i\omega t'} \int_{-\infty}^{\infty} dt F(t) e^{-i\omega t} = \frac{1}{2\pi} \int_{-\infty}^{\infty} dt F(t) \underbrace{\int_{-\infty}^{\infty} d\omega e^{i\omega(t'-t)}}_{2\pi \delta(t'-t)} = \int_{-\infty}^{\infty} dt F(t) \delta(t' - t) = F(t')$$

The above illustrates that the Dirac delta is a special function (distribution) for which:

$$\delta(x_0 - x) = \delta(x - x_0) = \begin{cases} 0, & \text{if } x_0 \neq x \\ \infty, & \text{if } x_0 = x \end{cases}, \text{ such that: } \int_{-\infty}^{\infty} \delta(x - x_0) dx = 1; \int_{-\infty}^{\infty} \delta(x - x_0) f(x) dx = f(x_0)$$

i.e. the Dirac delta is an infinitely narrow spike whose integral is 1.

More details in the *Mathematical methods in physics* course.

4

Supplement: proof of Fourier integral theorem (not very precise)

Let $F(t)$ be a continuous real function whose derivative is also continuous and whose absolute value is finite:

$$\int_{-\infty}^{\infty} |F(t)| dt < \infty$$

(It can be proven under stricter conditions, but that's all we're concerned with here.)

The inverse Fourier transform of the Fourier transform of $F(t)$: $\lim_{K \rightarrow \infty} \widetilde{F}_K(t')$, where:

$$\widetilde{F}_K(t') = \int_{-\infty}^{\infty} dt F(t) \underbrace{\int_{-K}^K d\omega \frac{e^{i\omega(t'-t)}}{2\pi}}_{\frac{\sin K(t'-t)}{\pi(t'-t)}} = \int_{-\infty}^{\infty} du F(t'-u) \frac{\sin Ku}{\pi u} = \int_{-\infty}^{\infty} du \underbrace{\frac{F(t'-u) - F(t')}{\pi u}}_{G(t',u)} \sin Ku + \frac{F(t')}{\pi} \int_{-\infty}^{\infty} du \frac{\sin Ku}{u}$$

This is a continuous, finite function of u , even at $u=0$ (here we get dF/du)

i.e.:

$$\lim_{K \rightarrow \infty} \widetilde{F}_K(t') = \lim_{K \rightarrow \infty} \underbrace{\int_{-\infty}^{\infty} du G(t',u) \sin Ku}_0 + F(t')$$

So the original function was indeed returned in the back-and-forth transformation!

5

Problems of real-world Fourier-analysis: finite measurement time

In real-world spectrum analysis we face two fundamental problems compared to mathematical Fourier analysis:

- (i) By nature, we can only measure for **finite time** instead of the infinite Fourier integral
- (ii) Instead of continuously measuring the signal, we can only **sample the data at discrete time intervals**.

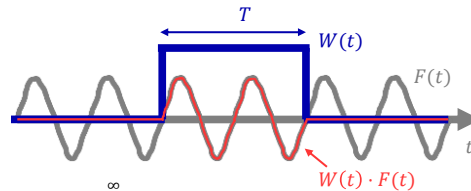
Let us investigate the consequences of (i) and (ii) separately.

First, let's assume that we sample continuously, but we measure for a finite T time.

The mathematical Fourier transformation:

$$f(\omega) = \int_{-\infty}^{\infty} dt F(t) e^{-i\omega t}$$

The finite measurement time is mathematically modeled by a **window function**, which is constant during the T time of the measurement, and zero otherwise ("rectangular window"). The Fourier transformation is calculated for the product of the $F(t)$ function and the $W(t)$ window function:



$$f_W(\omega) = \int_{-\infty}^{\infty} dt W(t) F(t) e^{-i\omega t}$$

Statement: the $f_W(\omega)$ Fourier transform of the product function $(W(t) * F(t))$ is practically the convolution of the $f(\omega)$ and $w(\omega)$ Fourier transforms of $F(t)$ and $W(t)$:

$$f_W(\omega) = \int_{-\infty}^{\infty} f(\omega') \cdot w(\omega - \omega') \frac{d\omega'}{2\pi}$$

Proof with Dirac-delta function:

$$\begin{aligned} f_W(\omega) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} dt dt' F(t) e^{-i\omega' t} \cdot W(t') e^{-i(\omega - \omega') t'} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dt dt' F(t) W(t') e^{-i\omega t'} \delta(t - t') = \\ &= \int_{-\infty}^{\infty} dt W(t) F(t) e^{-i\omega t} \end{aligned}$$

From the ω' integral

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Finite time continuous Fourier transformation of a harmonic function

$$F(t) = Ae^{i\omega_0 t}, \quad W(t) = \begin{cases} 1/T, & \text{if } |t| < T/2 \\ 0 & \text{otherwise} \end{cases}$$

The Fourier transform of $F(t)$ is a Dirac-delta:

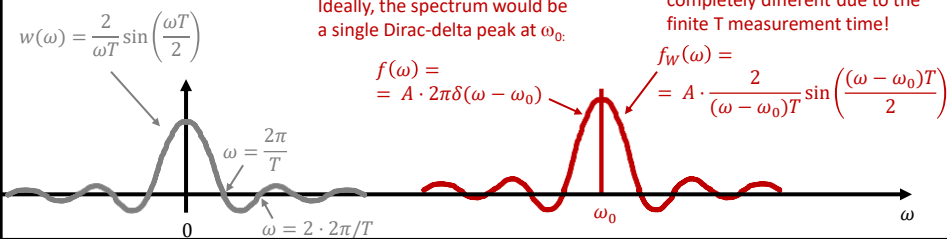
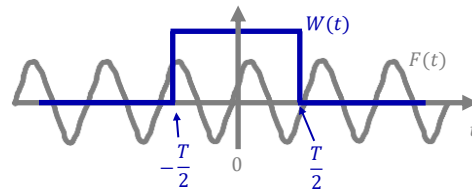
$$f(\omega) = A \cdot 2\pi\delta(\omega - \omega_0)$$

The Fourier transform of $W(t)$ is a $\sin(x)/x$ function:

$$w(\omega) = \int_{-\infty}^{\infty} dt W(t) e^{-i\omega t} = \frac{1}{T} \int_{-T/2}^{T/2} dt e^{-i\omega t} = \frac{2}{\omega T} \sin\left(\frac{\omega T}{2}\right)$$

The convolution of the two is the $\sin(x)/x$ function offset such that the central peak is shifted to ω_0 :

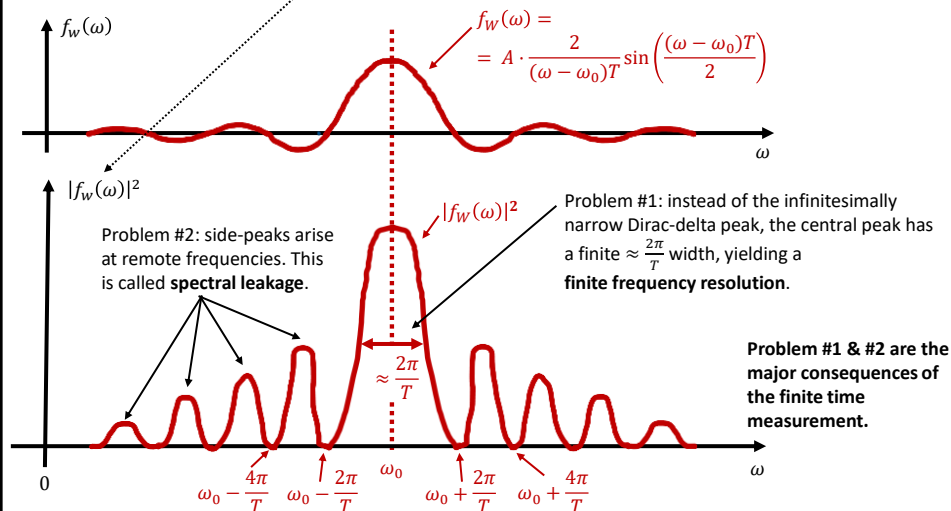
$$f_W(\omega) = \int_{-\infty}^{\infty} A \cdot 2\pi\delta(\omega' - \omega_0) \cdot w(\omega - \omega') \frac{d\omega'}{2\pi} = A \cdot w(\omega - \omega_0)$$



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Finite frequency-resolution and spectral leakage

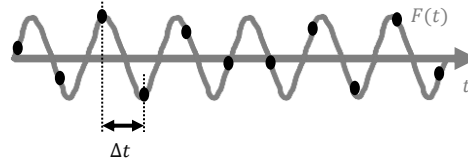
Further discussion of the previous slide: With a symmetric window function around the $t=0$ axis $w(\omega)$ is real, and thus $f_w(\omega)$ is also real. But at a general position of the window function a complex $f_w(\omega)$ is obtained. It is better to investigate $|f_w(\omega)|^2$, i.e. the "power" of the spectrum, which is independent of the position of the window function.



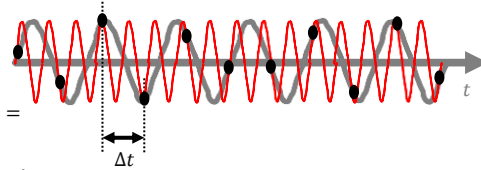
8

What, if we sample at finite discrete intervals, but for infinite time?

According to the **Nyquist-Shannon sampling theorem**: if a signal contains no components with frequencies higher than $f_{\max}$, then the signal can be perfectly described by its values sampled at intervals of $\Delta t = 1/(2f_{\max})$. Here $f_{\max} = 1/(2\Delta t)$ is called Nyquist frequency, or $\omega_{\max} = 2\pi f_{\max}$ is the Nyquist circular frequency.

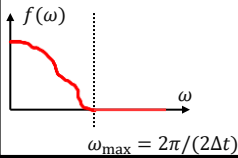


Aliasing: we sample a $F(t) = \sin(\omega t)$ signal at Δt intervals, $F_n = \sin(\omega n \Delta t)$. Problem: the same sample points are obtained for a much higher frequency signal, with $\omega' = \omega + 2k \cdot \omega_{\max}$, where k is an integer:



$$F_n = \sin((\omega + 2k \cdot \omega_{\max})n\Delta t) = \sin(\omega n \Delta t + kn2\pi) = \sin(\omega n \Delta t)$$

If our signal contains high frequency components with $\omega' > \omega_{\max}$, we cannot decide whether this high frequency signal, or the above $\omega < \omega_{\max}$ lower frequency signal is sampled.



If we have a band-limited signal, which does not contain any component at $\omega' > \omega_{\max}$, the finite Δt sampling is sufficient for the complete reconstruction of the full continuous spectrum, and the complete reconstruction of the temporal signal, once we collect infinite samples. Therefore, the discrete sampling with $N = \infty$ points yields a bandwidth limitation, i.e. the $f_{\max}$ Nyquist frequency limits our analysis, but the frequency resolution is not limited. The latter is related to the finite time of the measurement, and not the discrete sampling.

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Discrete Fourier transformation (DFT)

The Fourier integral with a time window of length T can be replaced by a discrete sum, where $F(n\Delta t)$ is the n^{th} datapoint of the $F(t)$ signal, which is sampled at Δt intervals, and altogether N datapoints are collected during the time of the measurement (T):

$$f_W(\omega) = \int_{-\infty}^{\infty} dt W(t) F(t) e^{-i\omega t} \Rightarrow f_W(\omega) \approx \sum_{n=0}^{N-1} W(n\Delta t) F(n\Delta t) e^{-i\omega n \Delta t} \cdot \Delta t, \quad T = N\Delta t$$

N is taken as an even number! (We'll come back to that later.) Note, that the DFT includes both the finite measurement time and the discrete sampling.

If the samples are taken at discrete time intervals, there is no reason to continuously evaluate the spectrum in the circular frequency domain ($f_W(\omega)$). Instead, we should evaluate the spectrum at discrete frequencies starting at $\omega = 0$ and using circular frequency steps of our approximate frequency resolution, $\Delta\omega \approx \frac{2\pi}{T}$:

$$f_W(\omega_k = k \cdot \Delta\omega) \approx \sum_{n=0}^{N-1} W(n\Delta t) F(n\Delta t) \frac{e^{-ik\Delta\omega n \Delta t}}{e^{-ikn2\pi/N}} \cdot \Delta t, \quad \text{Suppose we evaluate as many frequency points as there are sampled points (N):}$$

$$\omega_k = \frac{2\pi k}{N\Delta t}, \quad k = 0, 1, \dots, N-1$$

Symmetry:

$$f_W\left(\frac{(N-k)2\pi}{T}\right) = \sum_{n=0}^{N-1} W(n\Delta t) F(n\Delta t) e^{-\frac{i(N-k)n2\pi}{N}} \cdot \Delta t = \sum_{n=0}^{N-1} W(n\Delta t) F(n\Delta t) e^{+\frac{ikn2\pi}{N}} \cdot \Delta t = f_W^*(k \cdot 2\pi/T)$$

The $(N-k)^{\text{th}}$ point in the spectrum is the same as the complex conjugate of the k^{th} point! Accordingly, it is redundant to evaluate all the N points in the frequency space, it is enough to take the $\omega_k = \frac{2\pi k}{N\Delta t}$, $k = 0, 1, \dots, \frac{N}{2}$ points. These are altogether $N/2+1$ points. Note, that the first ($k=0$) and the last ($k=N/2$) points from these are real, and the rest are complex. This means that we have N real sampled datapoints in the time-space, from which 2 real and $N/2-1$ complex spectral values are calculated. The latter has the equivalent information content of N real numbers, i.e. the "information content is preserved". This makes sense!

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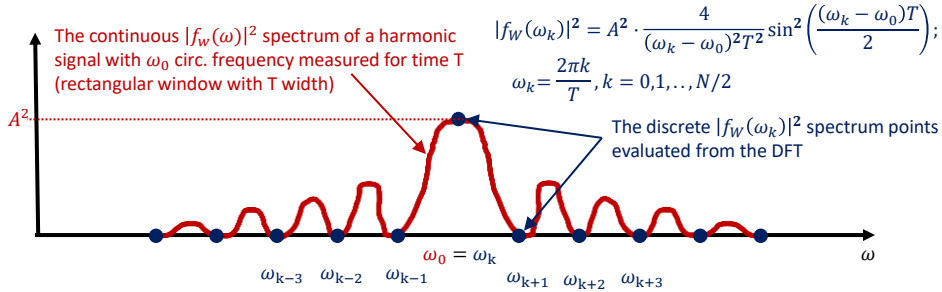
Discrete Fourier transformation (DFT)

Revisiting the DFT formula:

$$f_W(\omega_k) \approx \sum_{n=0}^{N-1} W(n\Delta t) F(n\Delta t) e^{-i\omega_k n\Delta t} \cdot \Delta t,$$

$$\omega_k = \frac{2\pi k}{N\Delta t}, \quad k = 0, 1, \dots, \frac{N}{2}$$

✓ $\omega_{\max} = 2\pi/(2\Delta t)$ Our DFT evaluation is in line with the sampling theorem.



In this example the discrete spectrum points are positioned at the zero values of $|f_W(\omega)|^2$ except for a single ω_k point, which is at the ω_0 circular frequency of the measured signal. In this case the value of the ω_k spectrum point gives A^2 , i.e. the amplitude of the signal can be directly read from the DFT. The $\omega_k = \omega_0$ condition yields:

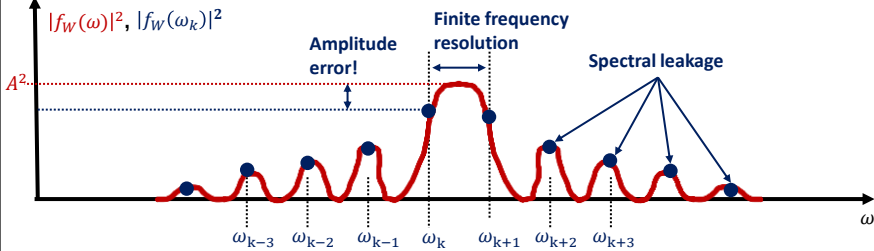
$$\omega_k = \frac{2\pi k}{T} = \omega_0 = \frac{2\pi}{T_0} \Rightarrow T = kT_0$$

This means, that the finite frequency resolution and the spectral leakage is not observed in the DFT, if the T measurement time is an integer multiple of the period of the signal.

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Discrete Fourier transformation (DFT)

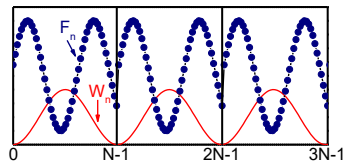
In a general case, we do not know the signal, so we cannot grant that the T measurement time is an integer multiple of the T_0 signal period. In this case the central DFT point is not at the peak of the real spectrum (ω_0), we cannot read the amplitude of the signal from the amplitude of the central DFT point, and the remote DFT points may catch the peaks of the spectral leakage.



A further property: the DFT of N datapoints is practically the same as the DFT of the periodic extension of the N datapoints:

$$f_W(\omega_k) = \sum_{n=0}^{N-1} W(n\Delta t) F(n\Delta t) e^{-i\omega_k(n+N)\Delta t} \cdot \Delta t = \sum_{m=n+N}^{2N-1} W(m\Delta t - N\Delta t) F(m\Delta t - N\Delta t) e^{-i\omega_k m\Delta t} \cdot \Delta t$$

If the time T does not equal a multiple of the period of the measured signal, the periodic extension causes the boundaries to jump abruptly. This can also explain spectral leakage, i.e. the appearance of components other than the fundamental frequency. This is helped by the disappearing window function at the edge of the time window.



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